

Band 9–10 Digital Technologies 2026 2?

Curriculum and assessment plan

St Brendan's College (Yeppoon)



Level description	Context and cohort considerations
By the end of Year 10 students should have had the opportunity to apply computational thinking by defining and decomposing real-world problems, creating user experiences, designing and modifying algorithms, and implementing them, including in an object-oriented programming language. Students use techniques, including interviewing stakeholders to develop user stories, to increase the precision of their problem definitions and solution specifications. They verify their solutions solve the problem by validating their algorithms, represented as flowcharts and pseudocode, and using test cases to confirm the correctness of their solutions. Students develop their object-oriented programming skills, and apply them to develop, modify and debug programs. They explain the importance of abstraction by representing online documents in terms of content, structure and presentation, as well as exploring simple data compression techniques and comparing their effectiveness. Students consolidate their skills in data acquisition and interpretation, cleaning and validating data to ensure it is accurate, consistent and domain appropriate. They model multidimensional data in more complex spreadsheets and relational databases, filtering and querying it to give insights into its meaning, and to pose further questions or make conclusions. They visualise this data in customisable ways, allowing greater exploration of trends and outliers to support or challenge their analyses. Students apply design thinking by using divergent techniques to generate design ideas for user experiences and solutions. They filter and prototype these ideas, developing user stories and applying design criteria based on current and future needs and enterprising opportunities, as well as their created user stories, and revise and further develop their preferred ideas based on their analysis. Students extend on these design criteria and user stories to evaluate the enterprise opportunities and future impact of existing solutions. Students consolidate their systems thinking by exploring how the hardware and software components of digital systems interact to manage, control and secure access to data. They increasingly use advanced features of existing and emerging digital tools to create interactive content for a diverse audience. They explore simple tools that help plan tasks, timelines and responsibilities for individual and collaborative projects. Students extend their knowledge of the importance of security by developing cyber security threat models and exploring an example of a supply chain vulnerability. They critique the digital footprint created by existing systems and their own solutions by applying the Australian Privacy Principles. In Digital Technologies, students should have frequent opportunities for authentic learning by making key connections to other learning areas.	The Years 9–10 cohort consists of a diverse group of boys with a wide range of digital competencies, from students with strong technical skills and personal interest in IT, to those who require further support to confidently engage with digital systems. Many students come from regional or rural backgrounds, where access to structured digital learning opportunities may have been limited. The program is designed to build on prior knowledge while introducing more advanced and applied skills relevant to future study and employment. The curriculum supports the school's priority to equip students with transferable digital skills, critical thinking abilities, and an understanding of ethical technology use in a rapidly changing world.

	Unit 1 — Websites 101	Unit 2 — Cyber Security Site	Unit 3 — Microbits - Data Gathering and Analysis	Unit 4 — Game Development
Year level	9	9	9	10
Duration	5 weeks	15 weeks	20 weeks	20 Weeks
Description	In this unit, students will explore the foundational technologies behind the internet and how websites function. They will examine how digital systems use hardware and software to manage and secure access to data across networks. Through guided investigation and hands-on tasks, students will learn how websites are constructed by separating content, structure, and presentation using HTML and CSS. They will also be introduced to simple data compression techniques and understand their importance in efficient web communication. This unit builds the knowledge and skills required for designing and analysing secure, well-structured, and efficient digital content online.	In this unit, students will investigate how cybercriminals design online scams and how these techniques can be identified and avoided. Through guided analysis and controlled, ethical simulations, students will explore common features of scam websites, including deceptive design choices, persuasive language, fake forms, and misleading links. By constructing mock scam websites in a safe learning environment, students will develop an understanding of how attackers exploit human behaviour rather than technical vulnerabilities. Students will also examine protective strategies such as recognising warning signs, verifying sources, and applying safe online practices. This unit builds students' awareness of cybersecurity risks and equips them with practical skills to analyse, evaluate, and respond safely to online threats.	In this unit, students will use micro:bit devices to collect real-world data and investigate how digital systems acquire, store, and process information. Students will design algorithms using logical operators to control data collection and represent these algorithms using flowcharts and pseudocode. Collected data will be transferred to spreadsheets, where students will apply techniques to validate, manage, analyse, and visualise data. Through interactive analysis, students will identify trends and outliers, draw conclusions, and make predictions. This unit develops students' understanding of data-driven decision making and the role of algorithms in automated data collection systems.	In this unit, students will design, develop, and analyse digital games using emerging game development platforms. Students will explore how user experience, algorithms, and data-driven decision making influence gameplay and player engagement. Through hands-on development and analytical tasks, students will prototype interactive game systems, implement and debug modular programs, and evaluate game designs against user needs and design criteria. The unit also introduces students to the use of data analysis and logical reasoning to improve game balance and functionality, developing skills relevant to game design, software development, and interactive media industries.
	Assessment 1 — Web Exam	Assessment 2 — Cyber Security Site	Assessment 3 — Data Collection	Assessment 5 — GDevelop Game
Term	1	2	3	1
Week	Week 5	Week 8	Week 8	Week 9

	Unit 1 — Websites 101	Unit 2 — Cyber Security Site	Unit 3 — Microbits - Data Gathering and Analysis	Unit 4 — Game Development
Description	Students will complete a written exam that assesses their understanding of how websites function and how data is managed in networked digital systems. The exam will include questions on the roles of hardware and software in securing and controlling access to data, the structure of web documents using HTML and CSS, and basic data compression techniques.	In this assignment, students will design and prototype a simulated scam website to investigate how online scams exploit users and digital systems. Students will define the problem using stakeholder input and user stories, then generate and evaluate design solutions. They will represent the website using content, structure, and presentation, explore simple data compression techniques, and collect, model, and analyse data related to scam behaviours using spreadsheets or databases. Students will collaborate using agile project management tools to create interactive content, develop cyber security threat models, and evaluate solutions against design criteria. The assignment also requires students to apply the Australian Privacy Principles to examine data misuse and propose strategies to protect digital footprints.	In this assessment, students will design and implement a micro:bit program to collect sports science–related data such as movement, acceleration, reaction time, or step counts. Students will represent their data collection algorithm using flowcharts or pseudocode and validate the program by testing outputs against expected results. Collected data will be transferred into a spreadsheet, where students will store, validate, analyse, and visualise the data to identify trends and outliers. Using this analysis, students will draw conclusions and make predictions about performance, demonstrating how algorithms and data analysis are used to measure and evaluate physical activity in sports science contexts.	In this assignment, students will design and develop an original digital game using GDevelop. Students will prototype the user experience of their game by designing gameplay mechanics, controls, feedback systems, and visual elements for a target audience. They will generate, modify, and communicate alternative design ideas, refining their game through iterative development. Students will implement, modify, and debug modular programs using selected algorithms and event-based logic to control gameplay behaviour. The completed game will be evaluated against design criteria and user stories, with students reflecting on usability, engagement, potential future impact, and opportunities for further development. Students will also use advanced features of GDevelop to create and communicate interactive content suitable for a diverse audience.
Technique	Examination	Project	Investigation	Project
Mode	Written	Multimodal	Multimodal	Multimodal
Conditions	- up to 50 minutes plus 10 minutes planning time - under supervised conditions - short response up to 100 words per item, up to 600 words for the task - extended response 200–300 words per item, up to 600 words for the task	4–6 A3 pages or equivalent digital media pages that may include annotated graphical representations. 1 minute presentation of the final product.	4–6 A3 pages or equivalent digital media pages that may include annotated graphical representations	4–6 A3 pages or equivalent digital media pages that may include annotated graphical representations
Achievement standard	By the end of Year 10 students develop and modify innovative digital solutions, decompose real-world problems, and critically evaluate alternative solutions against stakeholder elicited user stories. Students acquire, interpret and model complex data with databases and represent documents as content, structure and presentation. They design and validate algorithms and implement them, including in an object-oriented programming language. Students explain how digital systems manage, control and secure access to data; and model cyber security threats and explore a vulnerability. They use advanced features of digital tools to create interactive content, and to plan, collaborate on, and manage agile projects. Students apply privacy principles to manage digital footprints.	By the end of Year 10 students develop and modify innovative digital solutions, decompose real-world problems, and critically evaluate alternative solutions against stakeholder elicited user stories. Students acquire, interpret and model complex data with databases and represent documents as content, structure and presentation. They design and validate algorithms and implement them, including in an object-oriented programming language. Students explain how digital systems manage, control and secure access to data; and model cyber security threats and explore a vulnerability. They use advanced features of digital tools to create interactive content, and to plan, collaborate on, and manage agile projects. Students apply privacy principles to manage digital footprints.	By the end of Year 10 students develop and modify innovative digital solutions, decompose real-world problems, and critically evaluate alternative solutions against stakeholder elicited user stories. Students acquire, interpret and model complex data with databases and represent documents as content, structure and presentation. They design and validate algorithms and implement them, including in an object-oriented programming language. Students explain how digital systems manage, control and secure access to data; and model cyber security threats and explore a vulnerability. They use advanced features of digital tools to create interactive content, and to plan, collaborate on, and manage agile projects. Students apply privacy principles to manage digital footprints.	By the end of Year 10 students develop and modify innovative digital solutions, decompose real-world problems, and critically evaluate alternative solutions against stakeholder elicited user stories. Students acquire, interpret and model complex data with databases and represent documents as content, structure and presentation. They design and validate algorithms and implement them, including in an object-oriented programming language. Students explain how digital systems manage, control and secure access to data; and model cyber security threats and explore a vulnerability. They use advanced features of digital tools to create interactive content, and to plan, collaborate on, and manage agile projects. Students apply privacy principles to manage digital footprints.
Moderation	A sample of student portfolios representing a range of achievement levels will be reviewed by another Digital Technologies teacher, who will provide feedback on the consistency and fairness of the grading. Any necessary adjustments will be made to ensure alignment with the assessment criteria and standards.	Monitoring: Week 5 Conferencing: Week 7	Monitoring: Week 5 Conferencing: Week 7	Monitoring: Week 5 Conferencing: Week 7
			Assessment 4 — Wearable Sports Science	Assessment 6 — Rec Room VR - Game Analyst
Term			4	2

	Unit 1 — Websites 101	Unit 2 — Cyber Security Site	Unit 3 — Microbits - Data Gathering and Analysis	Unit 4 — Game Development
Week			Week 9	Week 9
Description			In this assignment, students will design and develop a wearable sports science data collection device using a micro:bit, along with a matching user interface to display or manage collected data. Students will define the problem by identifying user needs and developing user stories, then design and prototype the user experience of the wearable system. They will generate, refine, and communicate alternative design solutions, and evaluate their final solution against design criteria, user stories, and potential future improvements. Throughout the project, students will collaborate using digital tools and apply agile project management techniques to plan tasks, manage responsibilities, and address risks. This assignment develops students' ability to design user-focused digital systems and manage collaborative technology projects effectively.	In this assessment, students will act as gameplay analysts for a VR shooter prototype developed in Rec Room. Students will collect and analyse gameplay data to identify balance issues related to difficulty, player performance, and engagement. Using spreadsheets, students will visualise data, identify trends and outliers, and draw evidence-based conclusions about why players may disengage. Students will design and explain algorithms to improve enemy AI behaviour and gameplay balance, and propose a structured data model for a future high-score or player statistics system. Findings and recommendations will be presented in a written technical report supported by data analysis, demonstrating how quantitative evidence can be used to inform game design decisions rather than subjective judgement.
Technique			Project	Project
Mode			Multimodal	Multimodal
Conditions			4–6 A3 pages or equivalent digital media pages that may include annotated graphical representations	4–6 A3 pages or equivalent digital media pages that may include annotated graphical representations
Achievement standard			By the end of Year 10 students develop and modify innovative digital solutions, decompose real-world problems, and critically evaluate alternative solutions against stakeholder elicited user stories. Students acquire, interpret and model complex data with databases and represent documents as content, structure and presentation. They design and validate algorithms and implement them, including in an object-oriented programming language. Students explain how digital systems manage, control and secure access to data; and model cyber security threats and explore a vulnerability. They use advanced features of digital tools to create interactive content, and to plan, collaborate on, and manage agile projects. Students apply privacy principles to manage digital footprints.	By the end of Year 10 students develop and modify innovative digital solutions, decompose real-world problems, and critically evaluate alternative solutions against stakeholder elicited user stories. Students acquire, interpret and model complex data with databases and represent documents as content, structure and presentation. They design and validate algorithms and implement them, including in an object-oriented programming language. Students explain how digital systems manage, control and secure access to data; and model cyber security threats and explore a vulnerability. They use advanced features of digital tools to create interactive content, and to plan, collaborate on, and manage agile projects. Students apply privacy principles to manage digital footprints.
Moderation			Monitoring Week 5 Conferencing Week 7	Monitoring: Week 5 Conferencing: Week 7
	Unit 5 — Robotics			
Year level	10			
Duration	20 Weeks			
Description	In this unit, students explore the principles of Industry 4.0 through the design, development, and evaluation of autonomous robotic systems. Students investigate how hardware and software interact to control physical systems, using sensors, actuators, and algorithms to solve real-world automation problems. Through practical robotics projects, students design and implement modular programs, develop and validate control			

	Unit 5 — Robotics
	algorithms, and analyse sensor data to optimise system performance. The unit emphasises engineering design processes, systems thinking, and data-driven decision making, preparing students for pathways in robotics, automation, mechatronics, and industrial engineering.
	Assessment 7 — The Smart Warehouse Solution
Term	3
Week	Week 9
Description	In this assessment, students act as Automation Engineers tasked with designing a prototype Autonomous Guided Vehicle (AGV) for a smart warehouse logistics environment. Students will decompose the complex problem of autonomous warehouse operation into manageable sub-problems, object manipulation, and grid-based navigation. Using an object-oriented programming language (MicroPython), students will implement, modify, and debug a modular control system that governs the robot's behaviour. The robot must operate autonomously using a finite state machine to manage transitions between behaviours such as searching, lifting, and returning. Students will apply algorithms to interpret sensor data and calculate movement within a coordinate grid, demonstrating how hardware and software can be integrated to solve an authentic logistics challenge.
Technique	Project
Mode	Multimodal
Conditions	4–6 A3 pages or equivalent digital media pages that may include annotated graphical representations
Achievement standard	By the end of Year 10 students develop and modify innovative digital solutions, decompose real-world problems, and critically evaluate alternative solutions against stakeholder elicited user stories. Students acquire, interpret and model complex data with databases and represent documents as content, structure and presentation. They design and validate algorithms and implement them, including in an object-oriented programming language. Students explain how digital systems manage, control and secure access to data; and model cyber security threats and explore a vulnerability. They use advanced features of digital tools to create interactive content, and to plan, collaborate on, and manage agile projects. Students apply privacy principles to manage digital footprints.
Moderation	Monitoring: Week 5 Conferencing: Week 7
	Assessment 8 — The Systems Control Folio
Term	4
Week	Week 9
Description	In this assessment, students act as Control Systems Engineers responsible for testing, calibrating, and optimising their autonomous robot. Students will investigate how sensors and motors interact with software by designing and validating control algorithms, such as proportional (P) or PID control, to improve navigation accuracy and system stability. Students will conduct repeated test runs, collect sensor telemetry data, and analyse this data using graphs to evaluate the effectiveness of their control logic. The folio will document algorithm validation through before-and-after testing, showing how adjustments improve performance. Students will also include flowcharts representing their robot's finite state machine and provide a written evaluation of the system's limitations, reliability, and readiness for use in an enterprise context.
Technique	Investigation
Mode	Multimodal
Conditions	4–6 A3 pages or equivalent digital media pages that may include annotated graphical representations
Achievement standard	By the end of Year 10 students develop and modify innovative digital solutions, decompose real-world problems, and critically evaluate alternative solutions against stakeholder elicited user stories. Students acquire, interpret and model complex data with databases and represent documents as content, structure and presentation. They design and validate algorithms and implement them, including in an object-oriented programming language. Students explain how digital systems manage, control and secure access to data; and model cyber security threats and explore a vulnerability. They use advanced features of digital tools to create interactive content, and to plan, collaborate on, and manage agile projects. Students apply privacy principles to manage digital footprints.
Moderation	Monitoring: Week 5 Conferencing: Week 7

Content descriptions	Units					Content descriptions	Units				
Knowledge and understanding	1	2	3	4	5	Processes and production skills	1	2	3	4	5
Digital systems investigate how hardware and software manage, control and secure access to data in networked digital systems (AC9TDI10K01)	✓					Acquiring, managing and analysing data develop techniques to acquire, store and validate data from a range of sources using software, including spreadsheets and databases (AC9TDI10P01)		✓	✓	✓	✓
Data representation represent documents online as content (text), structure (markup) and presentation (styling) and explain why such representations are important (AC9TDI10K02)	✓	✓			✓	analyse and visualise data interactively using a range of software, including spreadsheets and databases, to draw conclusions and make predictions by identifying trends and outliers (AC9TDI10P02)		✓	✓	✓	✓
investigate simple data compression techniques (AC9TDI10K03)	✓	✓			✓	model and query entities and their relationships using structured data (AC9TDI10P03)		✓	✓	✓	✓
						Investigating and defining define and decompose real-world problems with design criteria and by interviewing stakeholders to create user stories (AC9TDI10P04)		✓	✓	✓	✓
						Generating and designing design algorithms involving logical operators and represent them as flowcharts and pseudocode (AC9TDI10P05)			✓	✓	✓
						validate algorithms and programs by comparing their output against a range of test cases (AC9TDI10P06)			✓	✓	✓
						design and prototype the user experience of a digital system (AC9TDI10P07)		✓	✓	✓	✓
						generate, modify, communicate and critically evaluate alternative designs (AC9TDI10P08)		✓	✓	✓	✓
						Producing and implementing implement, modify and debug modular programs, applying selected algorithms and data structures, including in an object-oriented programming language (AC9TDI10P09)				✓	✓
						Evaluating evaluate existing and student solutions against the design criteria, user stories, possible future impact and opportunities for enterprise (AC9TDI10P10)		✓	✓	✓	✓
						Collaborating and managing select and use emerging digital tools and advanced features to create and communicate interactive content for a diverse audience (AC9TDI10P11)		✓	✓	✓	
						use simple project management tools to plan and manage individual and collaborative agile projects, accounting for risks and responsibilities (AC9TDI10P12)		✓	✓	✓	
						Privacy and security develop cyber security threat models, and explore a software, user or software supply chain vulnerability (AC9TDI10P13)		✓			
						apply the Australian Privacy Principles to critique and manage the digital footprint that existing systems and student solutions collect (AC9TDI10P14)		✓			

General capabilities	Units				
	1	2	3	4	5
Critical and creative thinking	✓	✓	✓	✓	✓
Digital literacy	✓	✓	✓	✓	✓
Ethical understanding		✓			
Intercultural understanding					
Literacy	✓	✓		✓	✓
Numeracy				✓	✓
Personal and social capability		✓			

Cross-curriculum priorities	Units				
	1	2	3	4	5
Aboriginal and Torres Strait Islander histories and cultures					
Asia and Australia's engagement with Asia		✓			
Sustainability					

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